Packet Tracer - Verify Single-Area OSPFv2

Part 1: Verify the Existing OSPFv2 Network Operation

Step 1 – R1

Q: How did router R1 receive the default route?
A: R1 learned the default route through OSPF. In the routing table, it appears as **O*E2**, which indicates an external OSPF Type 2 route.

Q: From which router did R1 receive the default route?
A: The route was learned from **R2**, using **172.16.3.2** as the next-hop address through Serial0/0/0.

Q: How can you filter the output of `show ip route` to show only routes learned through OSPF?
A: Enter the following command:

```
None
R1# show ip route ospf
```

Q: Which routers have formed adjacencies with router R1?
A: R1 has established OSPF neighbor relationships with **R2 (2.2.2.2)** and **R3 (3.3.3.3)**.

Q: What are the router IDs and state of the routers shown in the command output?
A: The neighbor router IDs are **2.2.2.2 and 3.3.3.3**. Both neighbors are in the **FULL** state. Their priority is **0**, and the output shows `FULL/-` because the connections are point-to-point Serial links.

Q: Are all of the adjacent routers shown in the output?
A: Yes. R1 has two OSPF-enabled Serial connections, one to R2 and another to R3, so both OSPF neighbors appear in the neighbor table.

Q: Ping the ISP Router from PC1 — is it successful?
A: Yes. The ping works. One packet may be lost initially because of ARP address resolution, but connectivity to the ISP is established.

Step 2 – R2

Q: How did router R2 learn the default route to the ISP?
A: R2 has a static default route represented by *S 0.0.0.0/0**. This route points toward the ISP through G0/1 and is then advertised through OSPF.

Q: What type of OSPF network is attached to interface G0/0?
A: Interface G0/0 operates on a **BROADCAST** OSPF network.

Q: Are OSPF hello packets being sent out this interface? Explain.
A: No. G0/0 is configured as a **passive interface**, so OSPF does not transmit Hello packets through it. The LAN network can still be advertised by OSPF, but an OSPF neighbor relationship will not be established on this interface.

Q: Ping the S0/0/1 address on R3 from PC2 — is it successful?
A: Yes. PC2 can reach R3's S0/0/1 address successfully with **0% packet loss**.

Step 3 – R3

Q: Router R3 is routing for which networks?
A: R3 provides routing for **192.168.1.0/24**, **192.168.10.4/30**, and **192.168.10.8/30**, which are part of Area 0.

Q: What is the neighbor priority shown for the OSPF neighbor routers? Is this the default?
A: The neighbor priority is **0**. Yes, this is expected for the Serial point-to-point connections because these links do not perform a DR/BDR election.

Q: Ping the ISP Router from PC3 — is it successful?
A: Yes. PC3 successfully reaches the ISP router, with all four ping packets being received.

Part 2: Add the New Branch Office LAN to the OSPFv2 Network

Step 1 – Verify R4 Configuration

Q: Which interface is configured to not send OSPF update packets?
A: GigabitEthernet0/0/1 is configured as a passive interface. It connects to the Branch Office LAN, so OSPF updates and Hello messages are not exchanged through this interface.

Step 2 – Connect R4 to the OSPFv2 Network

Q: What state is displayed for router R3 (as seen from R4)?
A: R3 appears in the **FULL/DR** state. This means R3 has a complete OSPF adjacency and is acting as the Designated Router for the Ethernet network.

Q: Why is the state of router R4 different from the state of R1 and R2 (as seen from R3)?
A: The difference comes from the network types. The connection between R3 and R4 uses **Ethernet**, so OSPF performs a **DR/BDR election**. R3 becomes the DR and R4 becomes the BDR. R1 and R2 are connected through Serial point-to-point links, where DR/BDR election is not required, so their state is displayed as `FULL/-`.

Q: Ping PC2 from the Laptop — is it successful?
A: Yes. The Laptop can successfully communicate with PC2, receiving **4 out of 4 replies**. This confirms that the Branch Office LAN **192.168.11.0/24** has been added to the OSPF network and can be reached from the other LAN.